Urban Climate Adaptation Tools 🌍🌡️

A curated, open-source repository of publicly accessible tools and software for urban climate change adaptation. This collection helps planners, designers, community groups, and researchers find practical tools to build climate-resilient cities.

🎯 Purpose

Climate change is reshaping our cities through extreme heat, flooding, sea level rise, and other hazards. This repository catalogs free and publicly accessible tools that help communities understand, plan for, and adapt to these challenges.

📚 Tool Categories

🌡️ Heat Management

Tools for mapping urban heat islands, planning cooling strategies, and protecting vulnerable populations from extreme heat.

- 3 tools catalogued
- Focus: Temperature mapping, heat action plans, thermal comfort

💧 Flood & Water Management

Stormwater modeling, flood risk assessment, and green infrastructure planning tools.

- 4 tools catalogued
- Focus: Flood risk, drainage, water quality, green infrastructure

🌊 Coastal & Sea Level Rise

Tools for understanding coastal vulnerability, sea level rise impacts, and shoreline change.

- 3 tools catalogued
- Focus: Coastal flooding, erosion, storm surge, adaptation planning

🌳 Green Infrastructure & Nature-Based Solutions

Planning and analysis tools for urban forests, bioswales, green roofs, and other nature-based climate solutions.

- 4 tools catalogued
- Focus: Tree canopy, ecological design, ecosystem services

🏙️ Urban Planning & Vulnerability

Multi-hazard risk assessment and comprehensive adaptation planning platforms.

- 4 tools catalogued
- Focus: Risk assessment, vulnerability mapping, scenario planning

📊 Data & Mapping Platforms

Climate data repositories and mapping tools for spatial analysis.

- 4 tools catalogued
- Focus: Climate projections, GIS analysis, data visualization

⚡ Energy & Buildings

Tools for climate-adapted building design and energy resilience.

- 3 tools catalogued
- Focus: Building performance, passive design, energy modeling

🚀 Quick Start

1. **Identify Your Need**: What climate challenge are you addressing? (heat, flooding, sea level rise, etc.)
2. **Browse Category**: Navigate to the relevant category folder
3. **Review Tools**: Each tool has a detailed profile with use cases, requirements, and examples

4. **Check Guides**: Visit the guides folder for tutorials on choosing and using tools

🔍 Finding the Right Tool

Not sure where to start? Check out our decision guides:

- **Getting Started Guide** - New to climate adaptation tools
- **Choosing the Right Tool** - Match tools to your project needs
- **Data Sources Guide** - Where to find climate and geographic data

📋 Tool Profile Template

Each tool is documented with:

- **Description**: What it does and who it's for
- **Access**: URL, cost model (free/freemium/demo)
- **Geographic Coverage**: Local, national, or global
- **Technical Requirements**: Browser-based, GIS required, coding needed, etc.
- **Data Needs**: What inputs are required
- **Key Features**: Main capabilities
- **Outputs**: What you get (maps, reports, data)
- **Use Cases**: Real-world applications
- **Learning Resources**: Tutorials, documentation, videos
- **Examples**: Screenshots or case studies

🤝 Contributing

We welcome contributions! You can help by:

- **Adding new tools**: Submit tools we haven't catalogued
- **Updating existing profiles**: Improve descriptions or add examples
- **Sharing case studies**: Document how you've used these tools
- **Reporting issues**: Let us know about broken links or outdated info

- **Translating**: Help make this resource multilingual

See CONTRIBUTING.md for guidelines.

🌐 Tool Criteria

To be included in this repository, tools must be:

✅ **Publicly Accessible**: Free, freemium, or have meaningful free tiers/demos
✅ **Climate Adaptation Focused**: Directly support climate resilience planning
✅ **Urban Scale**: Applicable to cities, neighborhoods, or buildings
✅ **Actively Maintained**: Updated within the last 3 years
✅ **Documented**: Have user guides or documentation available

📖 Additional Resources

- **Case Studies**: Real-world examples of tools in action
- **Glossary**: Climate adaptation terminology
- **Funding Sources**: Grants for adaptation projects
- **Related Repositories**: Other useful GitHub repos
- **Academic Papers**: Research on adaptation tools

🏆 Featured Tools

Most Versatile: Climate Central Coastal Risk Screening Tool - global coverage, multiple scenarios, excellent visualization

Best for Beginners: NOAA Climate Explorer - intuitive interface, US coverage, great documentation

Most Comprehensive: Urban Climate Adaptation Knowledge Hub - hundreds of tools and resources

Community Favorite: i-Tree Suite - widely adopted, quantifies real benefits, excellent support

📊 Repository Stats

- **Total Tools**: 25+

- **Categories**: 7
- **Geographic Coverage**: Global
- **Languages**: English (translations welcome!)
- **Last Updated**: January 2026

📬 Contact & Community

- **Issues**: Report problems or suggest tools via GitHub Issues
- **Discussions**: Share experiences and ask questions in GitHub Discussions
- **Updates**: Watch this repository for new additions

📄 License

This repository is licensed under MIT License - the tool listings are informational and link to third-party resources with their own terms of use.

🙏 Acknowledgments

This repository builds on the work of:

- Climate adaptation practitioners worldwide
- Open-source software communities
- Organizations making tools publicly accessible
- Contributors who help maintain and expand this resource

⭐ **Star this repo** if you find it useful!

🔔 **Watch for updates** as we add more tools and resources.

🤝 **Contribute** to help others discover these valuable resources.